

MATH/QSS 30 – Spring 2026

(adapted from material by Olivia Chu and Brian Mintz)

Information about the Final Project

1 The Goals of the Final Project

The final project is meant to be a research project, which will require significant use of quantitative methods (e.g. mathematical modeling, statistical analyses, simulations) and concepts and ideas (e.g. Nash Equilibria, Evolutionarily Stable Strategies, replicator dynamics, network modeling) learned in class.

- Group Sizes are 4 maximum and randomly assigned.
- You will prepare a final report describing your findings (~10 pages not including references.), written in the format of a scientific paper with the following sections (for example):
 - Abstract
 - Significance Statement
 - Introduction
 - Results: Description of the model and analysis of the model
 - Data and Context (if applicable)
 - Discussion and Next Steps
- You will also give a flash talk (10-15 minutes) about your work to the class, in Week 9.

2 How to Choose a Topic

Work on something that you're interested in, something that you're excited to think about and explore. Think about how you can use evolutionary game theory to examine this question. The two types of styles of projects you can do are:

1. **“Analyze a game”**: look at an existing game and explore it fully – what are the winning strategies/Nash Equilibria/ESSes?
2. **“Apply game theory”**: choose one or more scenarios/situations and model them using game theory (more fun than option 1 in my opinion!).

Here are some tips on things to think about when choosing a topic:

- **Do some people watching.** Is there a phenomenon that you notice? Are there phenomena that you've always been curious about? Write down what you observe in these interactions, and see whether you can come up with a game describing them.
- **Consider different forms of heterogeneity**, at the individual level (e.g. different personality types) and the population level (e.g. in-group/out-group dynamics, different population structures like groups). How would the outcomes of games be different if we didn't interact in the same way as everyone else, but rather our interactions and how we interacted depended on these heterogeneities?

- **Take a look at the course readings** (even ones that we haven't gotten up to yet – read the abstracts or discussions/conclusions). Do any of the questions or modeling styles/tools interest you? See whether there are any ideas in the “Future Directions” section of the papers that you find interesting, and pursue one of them. Or, if you have other ideas about ways to extend the results (e.g. by looking at different parameters, making different assumptions in the model), explore that!
- **Think about the games we've studied in class** (e.g. Prisoner's Dilemma, public goods game, chicken, RPS) – how could we extend or tweak these games to make them more interesting, or to describe a certain real-world phenomenon? For example, an extension of RPS is RPSLS, or Rock Paper Scissors Lizard Spock, which Sheldon played on The Big Bang Theory.
- **Think about games or strategic interactions you might have seen in sports, TV shows, or movies.** For example, there is an evolutionary invasion of RPS on Friends.
- **Read social science papers**, where the authors do an experiment and/or analyze a certain dataset. Do the results of the studies seem generalizable to a larger population? Think about whether you could translate the experiment that they did into an evolutionary game theory problem (e.g. the way we did for the public goods game), where you could test the results using different setups and parameters.
- **Think about big problems that face us today**, like climate change/negotiations, political polarization, the relationship between medical providers and insurance companies or pharmaceutical companies, coalition building in politics (in the US, but also in other countries, where there are more than two dominant political parties), the spread of diseases, misinformation, propaganda in person and online. Can we think of any of these problems in terms of strategic interactions and payoffs?

3 Final Project Timeline

3.1 Week 3: Problem Set 1 Due

You're encouraged to use Problem Set 1 to brainstorm ideas for the final project (with your group if you'd like, if you already have one). You don't have to write out a whole mathematical model, of course, but the more details you give me about how you might be able to apply EGT to your problem, the more feedback and advice I can give you in return.

3.2 April 21: Final Project Proposal Due

Please submit one proposal per group. Don't worry if you don't exactly know all the details of how you'll try to answer your question. This will become clearer as we see more examples and learn more tools in class.

The minimum is one page not including references. The format is single space with 12pt font arial and apa citations. Think about the following:

- **Motivation for pursuing the problem** – why are you interested in this problem, and why is it important?
- **Background information** – what are the key ideas that we (the audience) should know?
- **Prior work done** – do you know of any research that has looked at problems similar to yours?
- **Application to EGT** – how do you propose using EGT to explore the problem?
- **Model setup** – how will you set up the problem mathematically? For example, is there a payoff matrix that you'd like to use? What parameters will you explore? What is the structure of individuals' interactions (i.e. who interacts with whom? Is it a network problem?)
- **Hypotheses** – what do you predict the results will be? What is your plan if the results are different from what you expected?
- **Tools you'll use** – how do you intend to use the tools we've learned in class to pursue your problem? (e.g. analyzing the replicator dynamics analytically, using software tools like Dynamo, doing stochastic simulations, analyzing a dataset, doing statistical analyses, analyzing networks/graphs)

3.3 Weeks 5–8: Working on the Final Project

The bulk of the work for the final project will be done during weeks 5–8 of the quarter. During these weeks, each group should meet with mentors weekly, to update them on your progress and ask any questions that come up. Meetings and their attendance will be your responsibility and graded.

4 Project Ideas

In addition to the recommendations in Section 2 above, here are some other places to look for project ideas:

- This website lists examples you can analyze or adapt from fiction/pop culture that involve game theory.

Games to analyze:

- Hackenbush
- Dots and Boxes
- Domineering
- Sprouts
- Traveler's dilemma